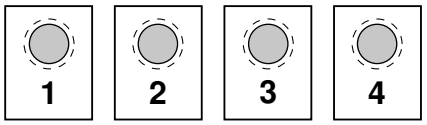
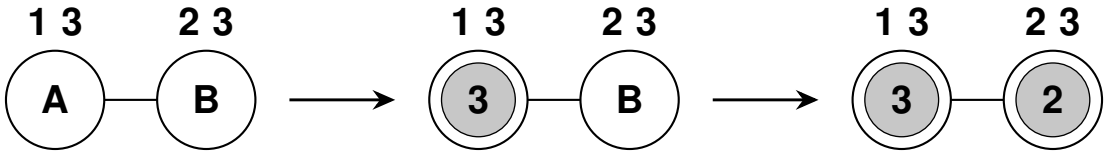
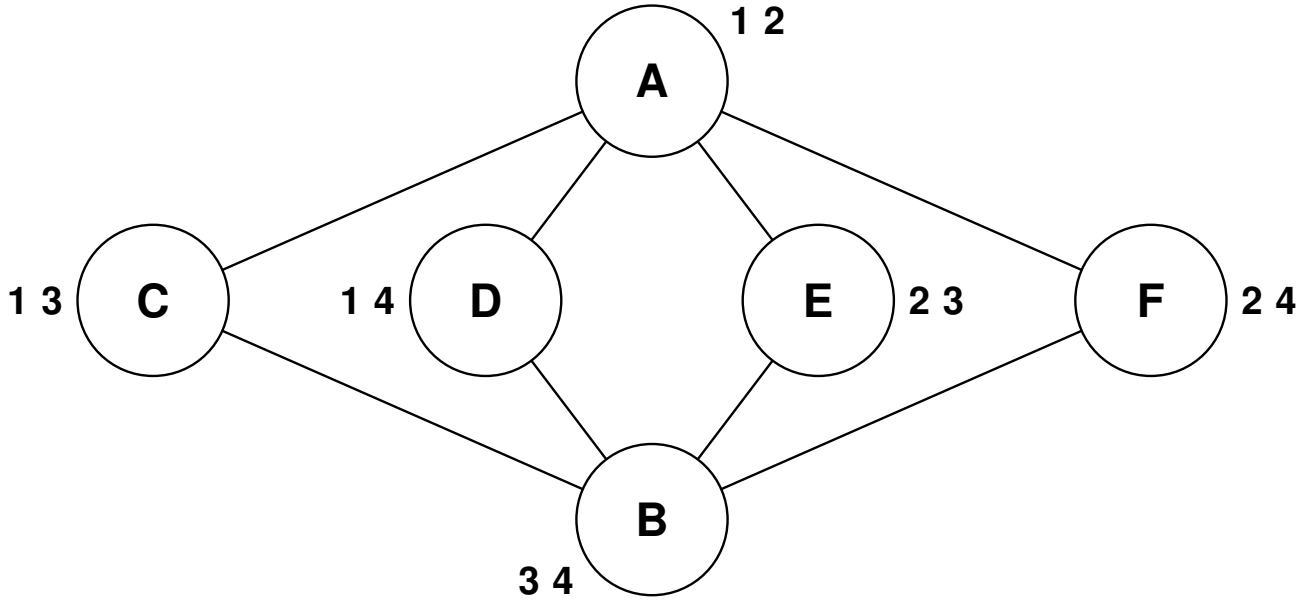
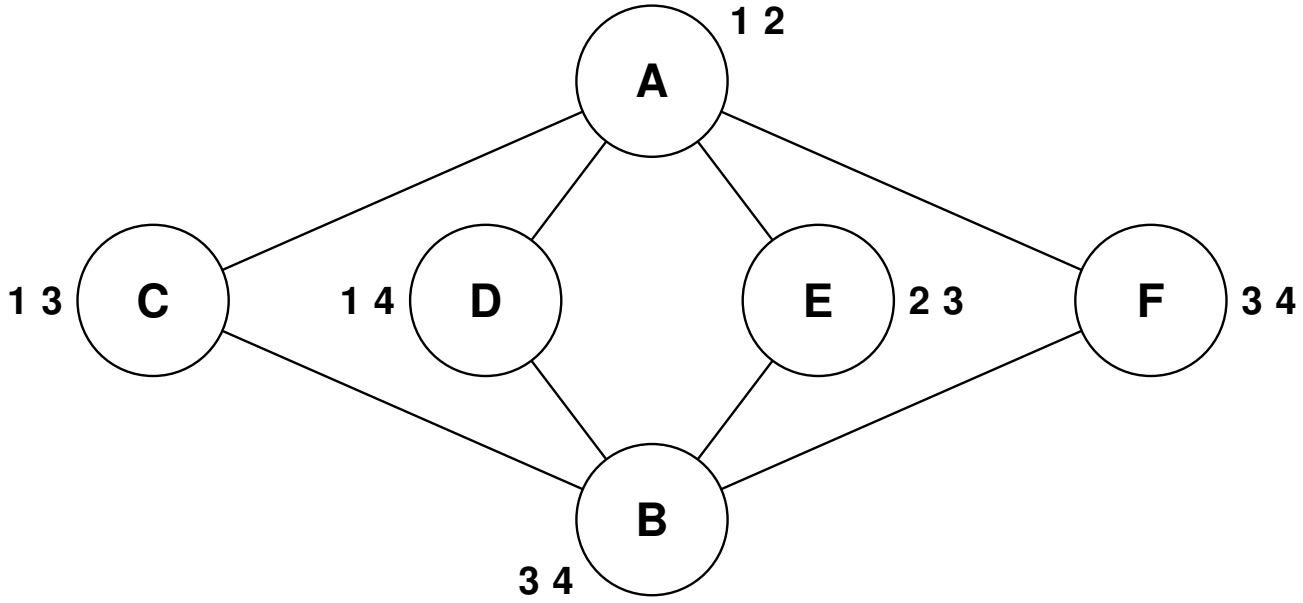
The headers show which colour goes with each number. Put one counter on each circle. Two circles joined by a line must not get the same number. When numbers are printed next to a circle, its counter must be one of them.



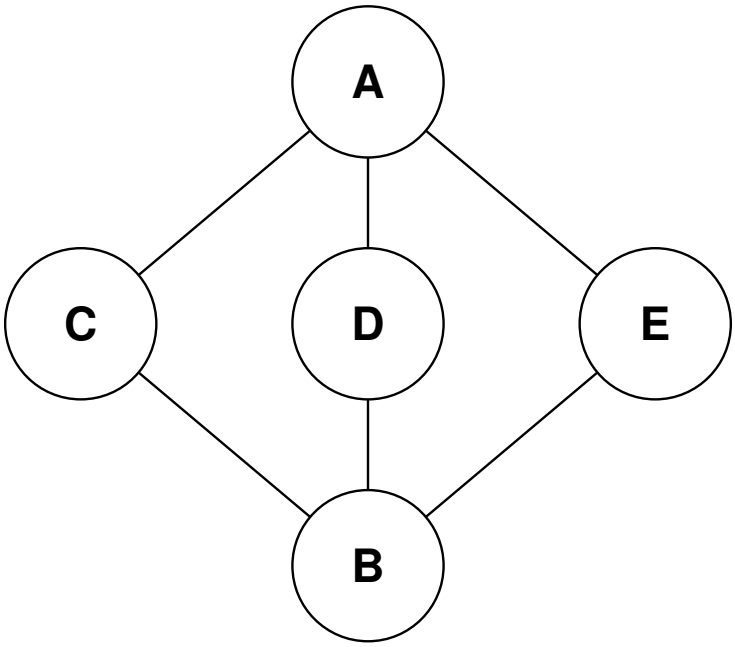


Problem 1: Which of these can be done? For any that can't, explain why.

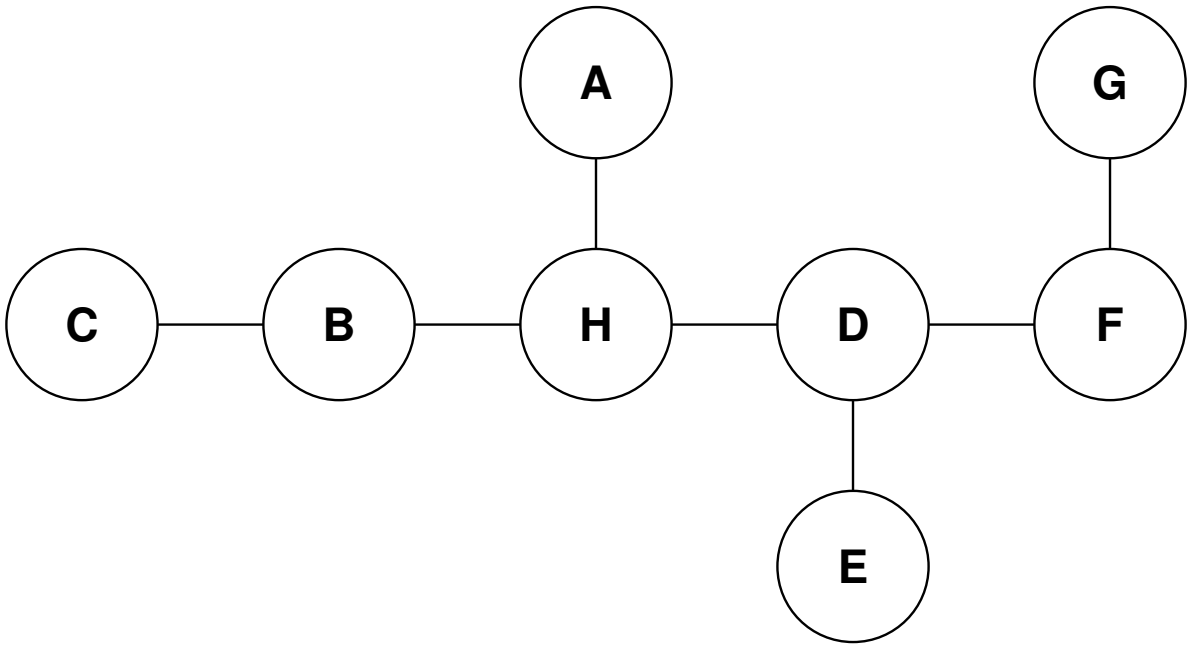




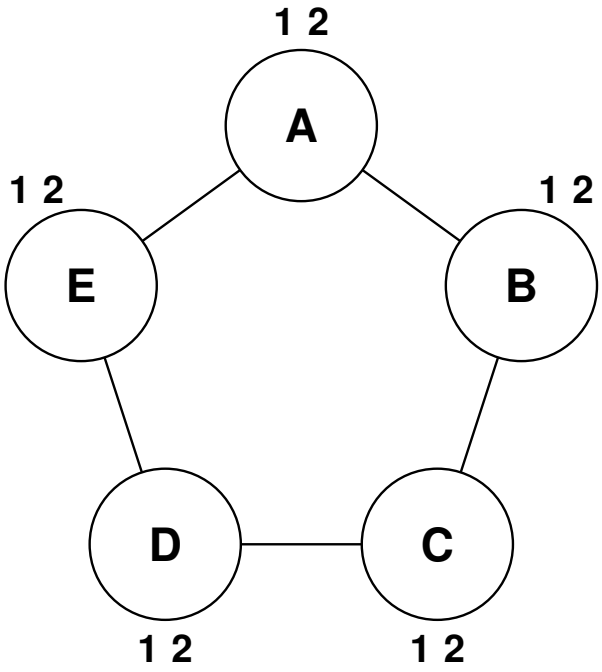
Problem 2: Can you write two of the numbers 1, 2, 3, 4 next to every circle of this network so that it can't be done? Explain.

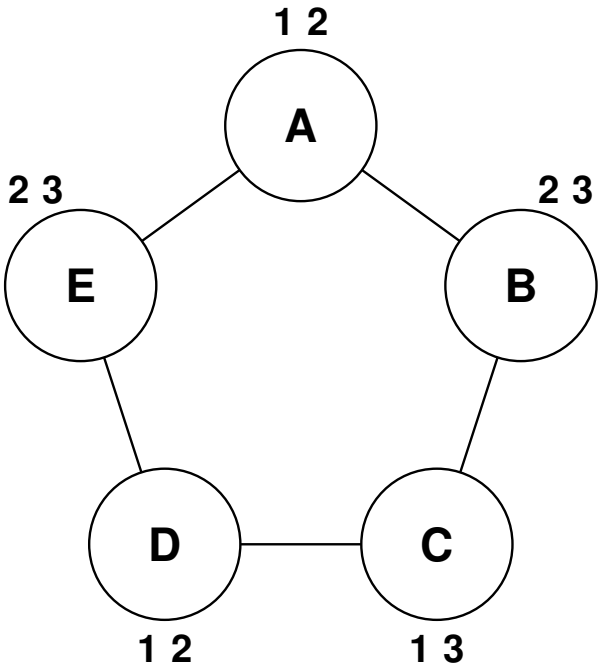


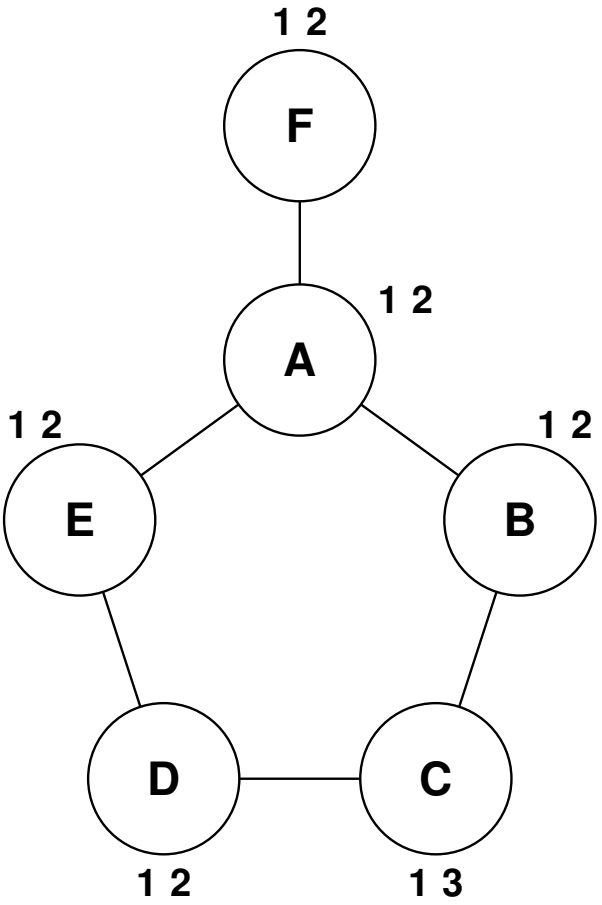
Problem 3: Can you write two of the numbers 1, 2, 3, 4 next to every circle of this tree so that it can't be done? Explain.

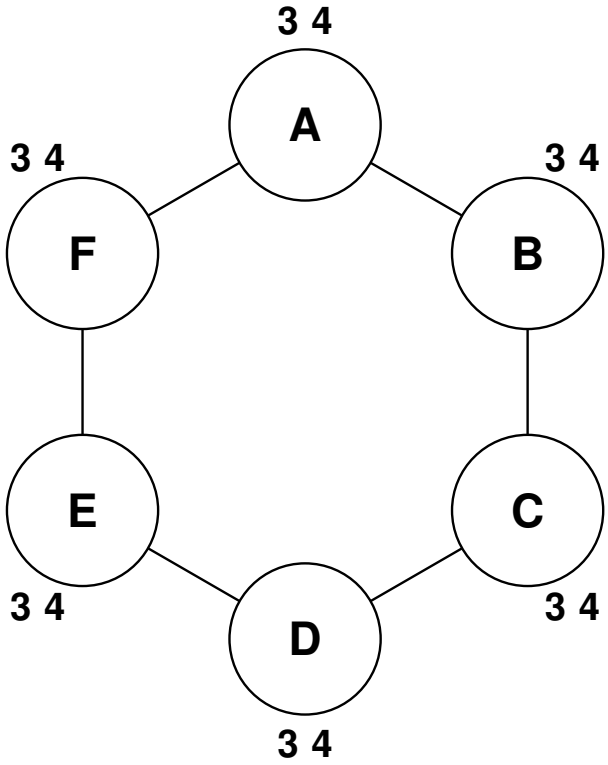


Problem 4: Which of these can be done? Suppose every circle of a ring, of any size, has two numbers next to it. When can it not be done? Explain.

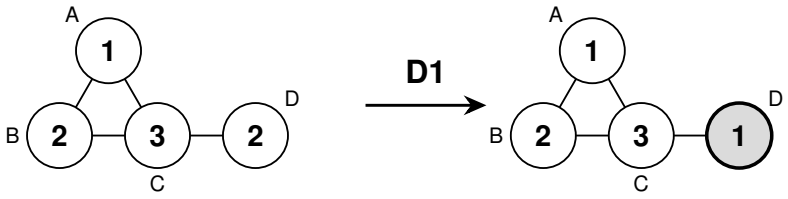




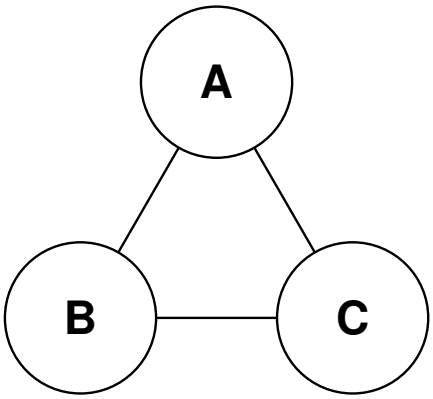




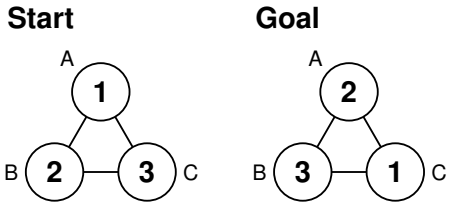
A move takes one counter off a circle and puts a counter with a different number there. After every move, no line may have the same number at both ends. Write a move as the circle's letter and its new number, like D1.



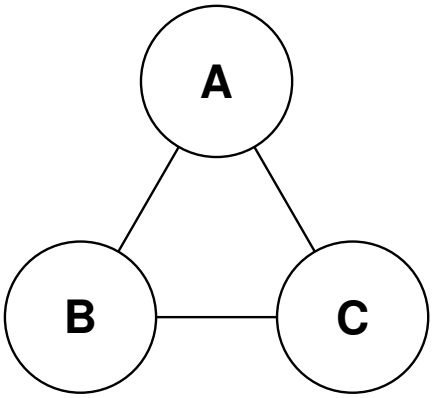
Problem 5: Using only the counters shown, can you change Start into Goal? Write the fewest moves you can find, or explain why it can't be done.



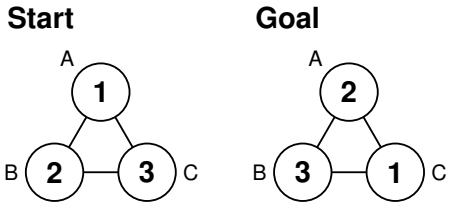
Counters 1 2 3



Moves: _____

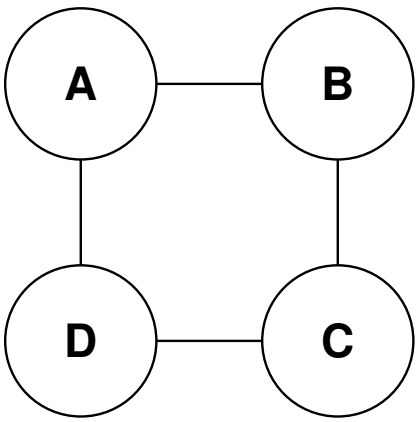


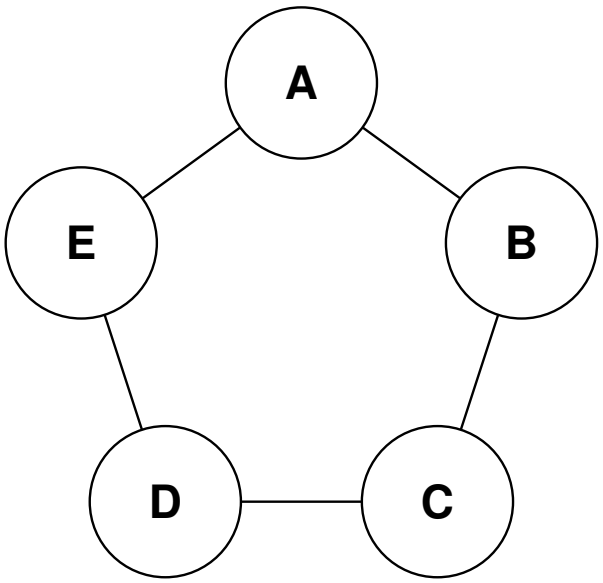
Counters 1 2 3 4

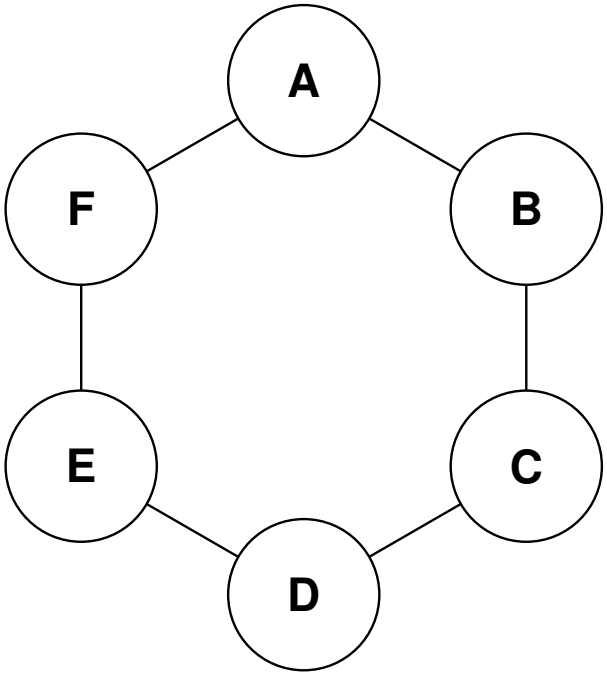


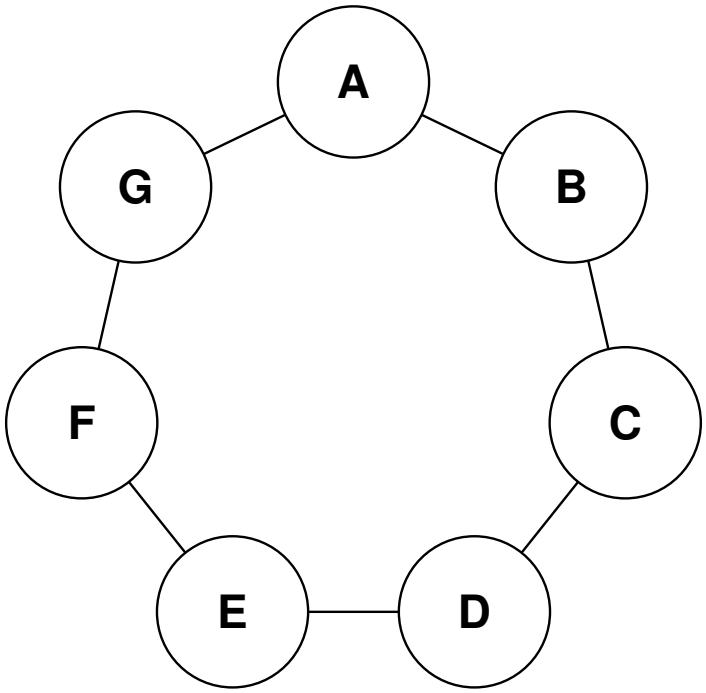
Moves: _____

Problem 6: Use counters 1, 2 and 3. On which of these rings can you place counters, with different numbers at the two ends of every line, so that no counter can move? Explain.

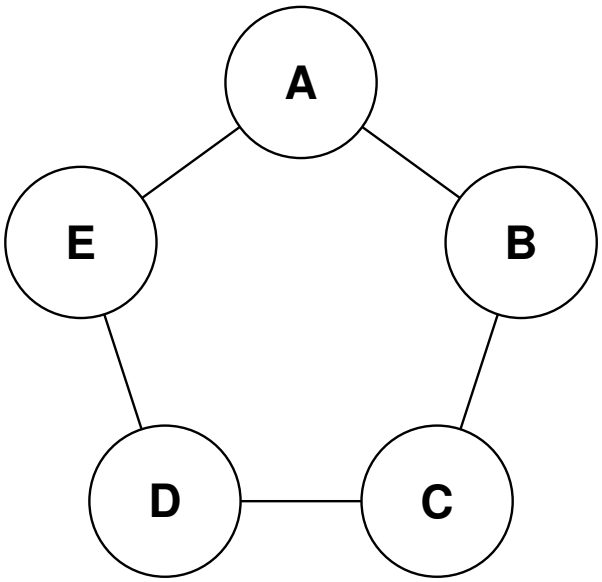




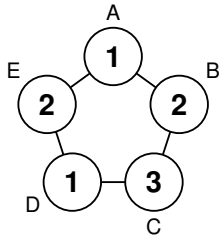




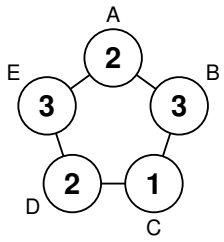
Problem 7: Use counters 1, 2 and 3, and begin at Start each time. Which goals can you reach? Write your moves. For a goal you can't reach, explain why.



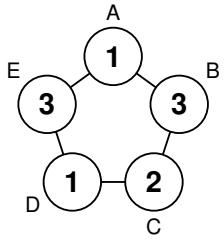
Start



Goal 1



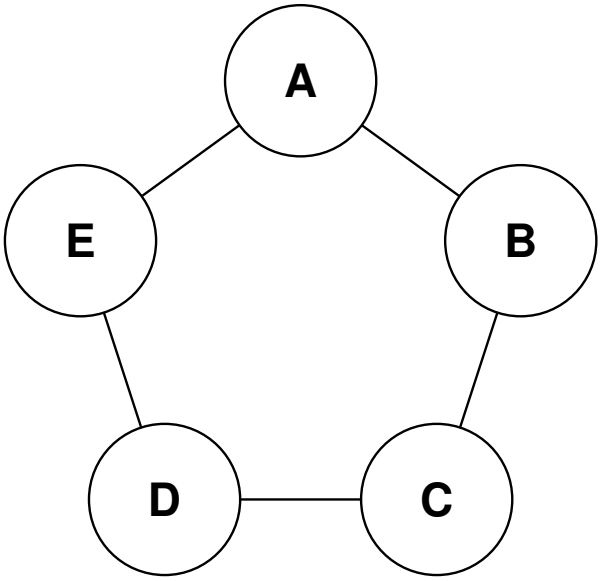
Goal 2



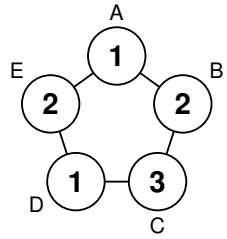
Goal 1: _____

Goal 2: _____

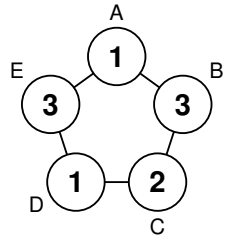
Problem 8: Now counter 4 may be used too. Change Start into Goal in as few moves as you can. On a ring of any size, with counters 1, 2, 3, 4, can every placement with different numbers at the two ends of every line be changed into every other one? Explain.



Start

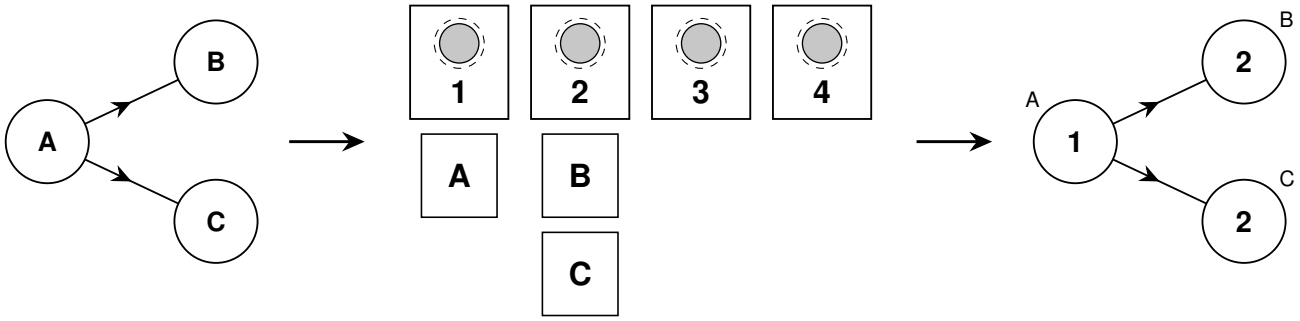


Goal

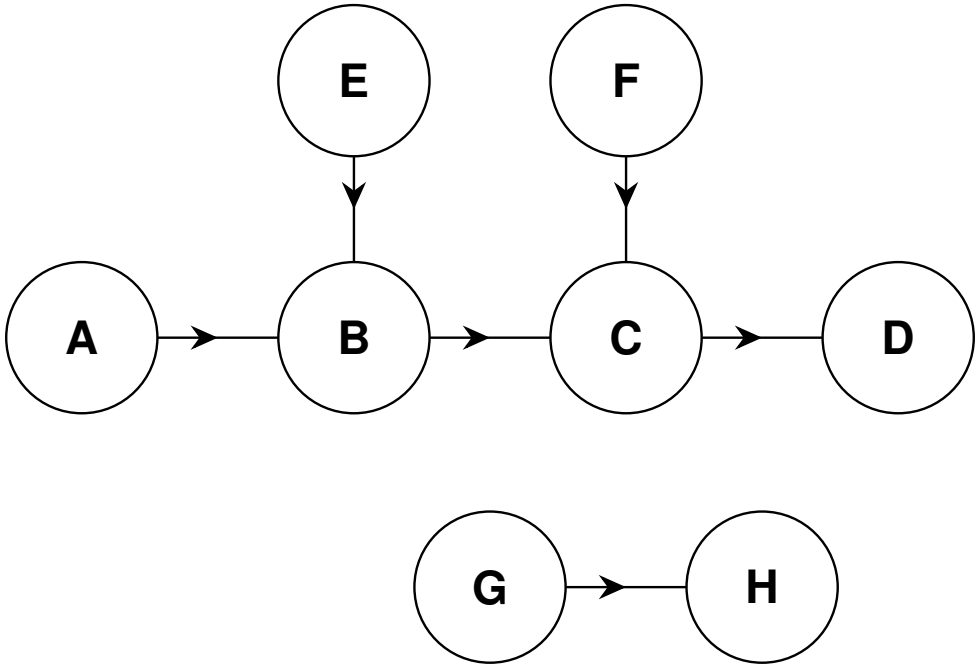


Moves: _____

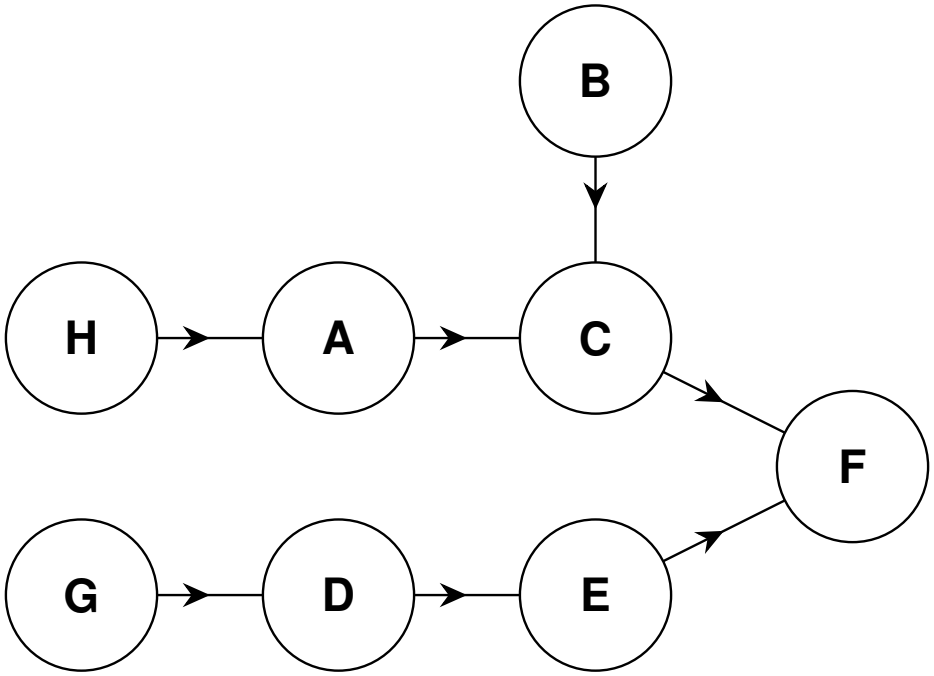
Lay the headers in a row, 1, 2, 3, 4, where 1 comes first. Put each card under a header, and a counter with that header's number on the card's circle. An arrow from one card to another means the first card must come earlier, under a smaller number. Cards with no arrow between them may share a header.

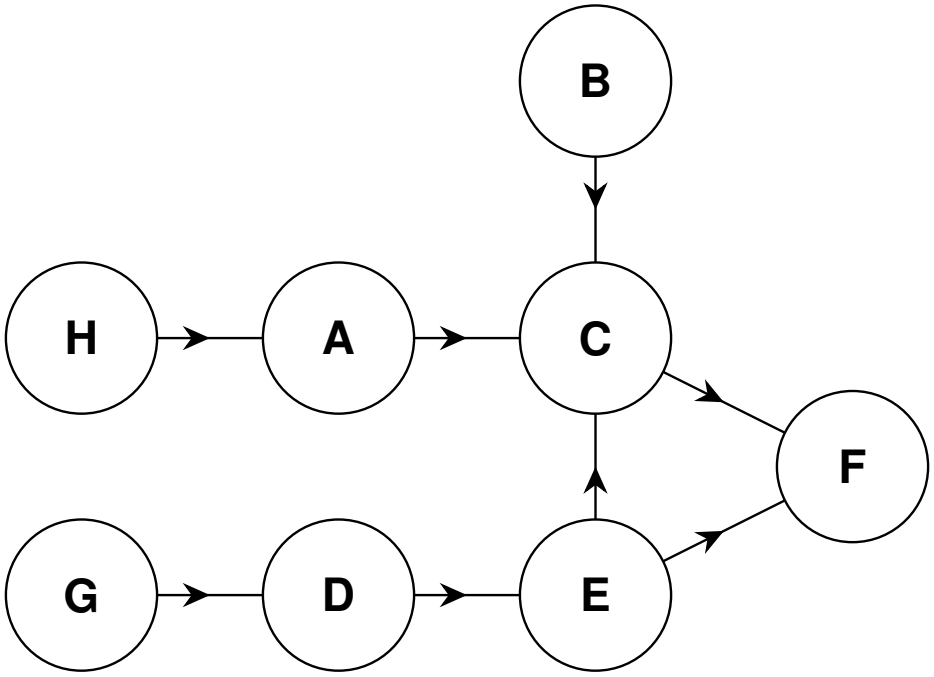
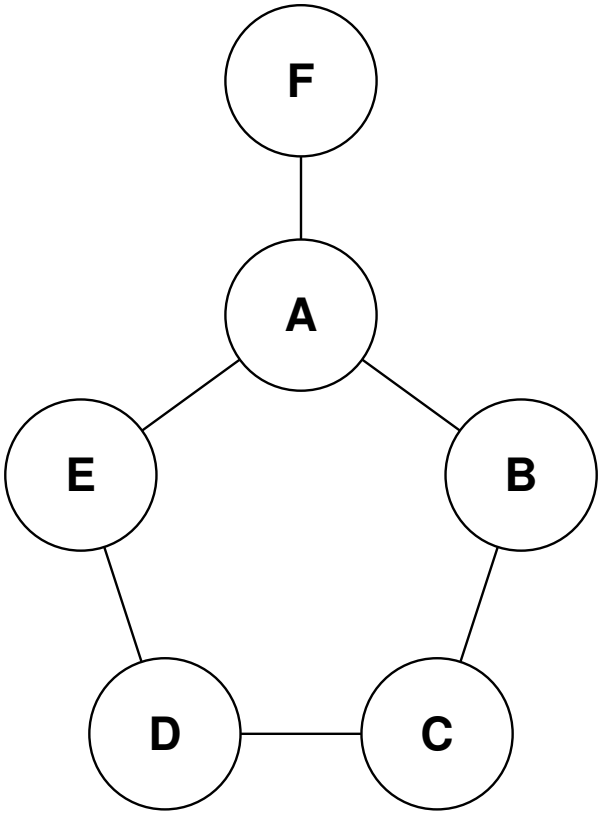
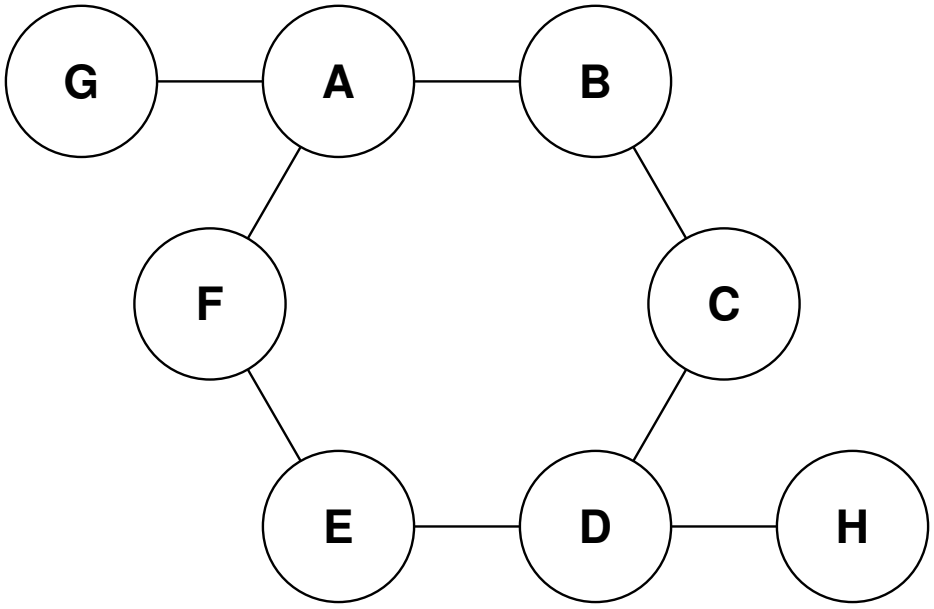


Problem 9: Put every card under a header, using as few headers as you can. Explain why fewer headers can't work.



Problem 10: Can the cards of each network be put under the four headers?





Problem 11: Draw an arrowhead on every line so that the longest path along the arrows passes through as few cards as you can. Then put the cards under headers to follow your arrows. On the first network, can you draw the arrows so that no path along the arrows passes through three cards? Explain.